PV-Finder: additional studies for ANA-IDTR-2026-01

Status of the reviewer-requested studies

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Overview

HL-LHC: 1 920 PU200 $t\bar{t}$ events (r16443, $185 \leq \mu \leq 200$) unless stated.

Study	Status	Result
1. Small model for CPU	Training	s5: $6.4\times$ faster on CPU
2. Matching window	Done	0.1694 mm for all vertices, 0.1 mm for HS
3. Split vertices in efficiency	Done	Identical efficiency; PV-Finder $+3.36$ points over AMVF
4. HS studies vs μ	Done	Clean HS: GNN 96.4%, AMVF 92.5%
5. AMVF HS selection vs ITk	Done	Stored AMVF primary: 92.9%, matches ITk
6. ITk HS resolution	Done	PV-Finder + GNN 9.3 µm, AMVF 10.3 µm
7. Run 2 vs Run 4 gap	Done	Not methodology; AMVF/PV-Finder ratio $1.93 \rightarrow 1.33$
8. Data–MC	Done	MC on all Fig. 6.2 panels; AMVF-reference bias $+2.4$ points
9. z_0 baseline in Fig. 7.2	Done	Nearest z_0 31.9%, AMVF 30.9%, GNN 68.4%
10. Timing vs μ	Input	Per-event Athena timing from integration
11. AMVF weights vs GNN	Answered	Compared against truth instead
12. Paper figures, Table 1	Done	Figs. 1–7 and Table 1

1. Smaller PV-Finder for CPU

Comment. Faster CPU inference without loss of performance.

What we did.

- Narrower U-Net (s5: 112 channels), same data and loss as v6, 1+1 epochs.
- Packed inference: only real tracks, no padding.

Why not alternatives.

- INT8 quantization: $\lesssim 1.25\times$.

Model	Params	CPU ms/event	Speed-up	ΔEff (points)
v6 production	3.55M	1601	1.0	–
v6, packed	3.55M	1159	1.4	0
s5, packed	0.67M	250	6.4	-0.33 / -0.26

CPU vs CPU, ONNX Runtime, 1 thread. s5: 0.8 fewer fakes per event.

1. Smaller PV-Finder for CPU: Athena time at $\mu = 200$

What we did.

- Note Athena CPU timing (Xeon Silver 4216) scaled by the measured speed-ups.

CPU setup	Inference (ms)	Full chain (ms)	vs AMVF (~650 ms)
v6 (note)	2189	2400	3.7× slower
v6, packed	1560	1780	2.7× slower
s5, packed	340	555	1.2× faster
s8, packed	245	455	1.4× faster

Full chain = inference + track selection (31 ms) + GNN (182 ms).

2. Matching window

Comment. Justify the window; use it everywhere.

What we did.

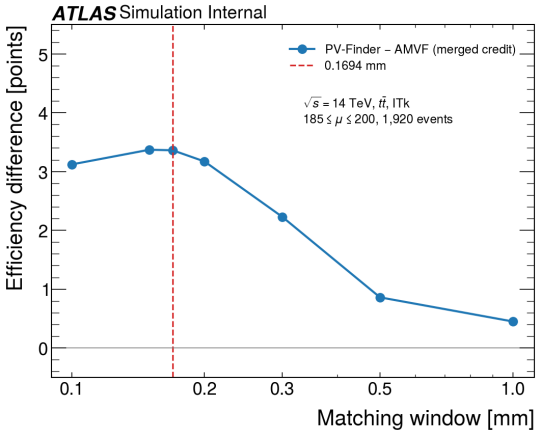
- 0.1694 mm $= 3\times$ the 68% position residual (56.45 µm); 0.1 mm for HS.
- Window scan on the same events.

Why not alternatives.

- $\sigma_{\mathrm{vtx-vtx}}$ (0.229 mm) measures vertex separation, not position.
- Wider windows add chance matches.

Result.

- 83.16% vs 79.80%: $+3.36 \pm 0.06$ points.
- PV-Finder ahead at every window.



3. Split vertices in the efficiency

Comment. Include split vertices in the efficiency.

What we did.

- Added a split-inclusive efficiency to the evaluation.
- Unit tests for every matching case.

Result.

- Identical to the standard efficiency: with closest-first matching, the truth vertex next to a split is already matched.
- PV-Finder 83.16%, AMVF 79.80%, $+3.36 \pm 0.06$ points.

4. Hard-scatter studies vs pile-up

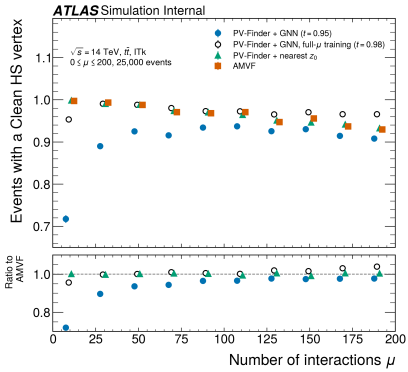
Comment. HS reconstruction, clean HS, HS tracks, HS selection vs μ.

What we did.

- Four HS metrics in 10 μ bins, 25 000 events; ATLAS definitions, same code for both.

Result.

- HS reco: 99.9% / 99.8%.
- Clean HS: GNN 96.4%, AMVF 92.5%.
- HS tracks on HS vertex: 74.9% / 74.1% (purity 91.3% / 88.4%).
- Reco + selected: 96.3% / 92.9%.



Clean: ≥70% of the vertex tracks from one interaction, not split (ATLAS Matched; tracks counted, not fit-weighted). HS vertex: most HS tracks. Purity: HS tracks / all tracks on that vertex. Selected: largest $\sum p_{\mathrm{T}}^2$.

5. AMVF HS selection vs the ITk paper

Comment. AMVF HS selection 80.7% vs ITk ~92%.

What we did.

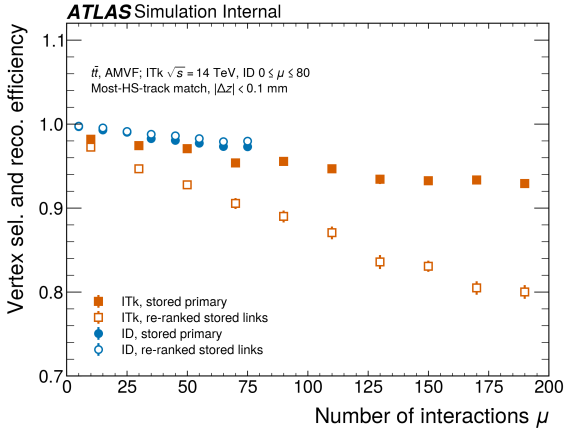
- Used the stored AMVF primary vertex.

Why not alternatives.

- Re-ranking stored track lists by $\sum p_{\mathrm{T}}^2$ picks a different vertex in $428/1920$ events.

Result.

- 92.9%, matches ITk.



6. Longitudinal HS resolution, ITk definition

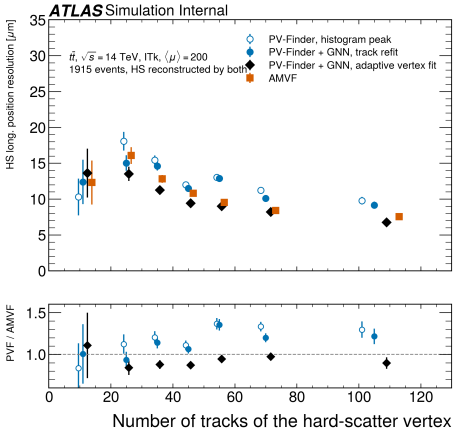
Comment. HS longitudinal resolution with the ITk definition.

What we did.

- Athena iterative RMS, HS within 0.1 mm, same 1 915 events.
- Adaptive vertex fit with beam-spot constraint on the GNN-assigned tracks.

Result.

- PV-Finder + GNN: 9.27 ± 0.25 µm.
- AMVF: 10.25 ± 0.30 µm (ITk: ≈ 10 µm).
- 1.0 µm better than AMVF; histogram peak alone 12.38 µm.



7. Run 2 vs Run 4 resolution gap

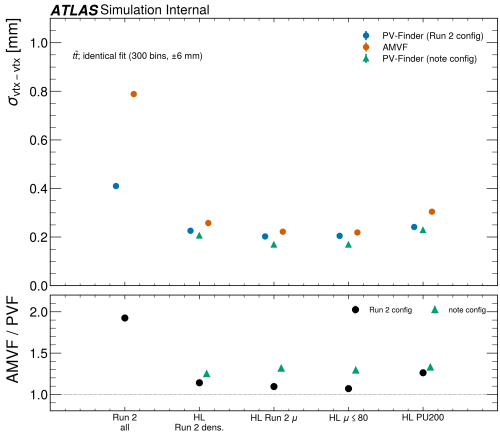
Comment. Why is the $\sigma_{\text{vtx}-\text{vtx}}$ gap larger in Run 2?

What we did.

- Same binning, range, statistics and selection for both.

Result.

- Run 2: 0.410 vs 0.789 mm, ratio 1.93.
- PU200: 0.229 vs 0.305 mm, ratio 1.33.
- Method choices move the ratio by ≤ 0.06.



8. Data–MC validation

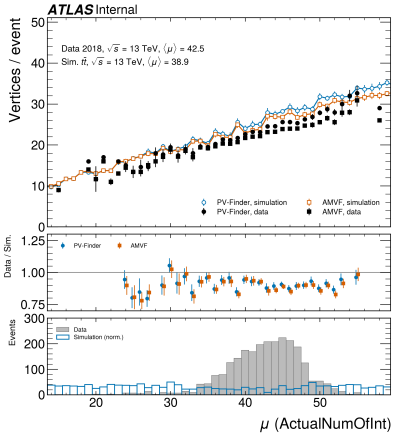
Comment. MC on all data plots; Run 3 MC; AMVF vs truth reference.

What we did.

- MC on all Fig. 6.2 panels.
- In MC, swapped truth for AMVF as reference.

Result.

- Run 2 $\sigma_{\text{vtx}-\text{vtx}}$: MC 0.416, data 0.368 mm.
- AMVF reference: $+2.36$ points recovery, $+1.30$ unmatched peaks per event.
- No Run 3 MC sample exists.



9. Track association: z_0 baseline in Fig. 7.2

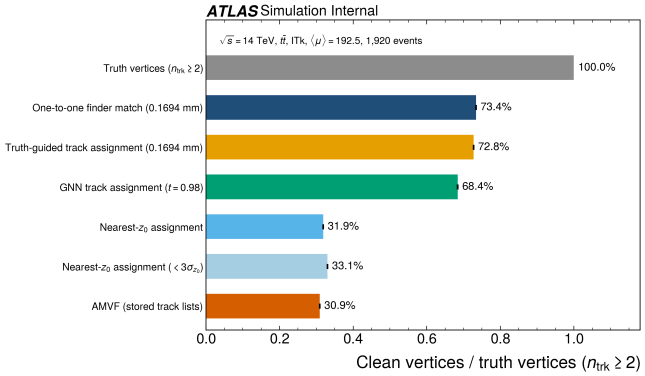
Comment. Compare the GNN with a z_0-based association.

What we did.

- Nearest vertex in z_0, and within $3\sigma(z_0)$.

Result.

- Finder limit 73.4%.
- GNN 68.4%.
- Nearest z_0 31.9%, 3σ 33.1%.
- AMVF 30.9%.



10. Timing vs pile-up

Comment. Timing vs pile-up for both algorithms.

What we did.

- Plotting code ready for per-event time vs μ, CPU and GPU.

Result.

- Input: per-event Athena timing (algorithm, device, μ, ms) from the integration team.
- Current note figure: PV-Finder on GPU faster than AMVF at all μ.

11. AMVF track weights vs GNN scores

Comment. Compare GNN track scores with AMVF track weights.

What we did.

- Compared both against truth: HS-track fraction and purity (study 4).

Why not alternatives.

- AMVF weights are not stored in the ntuples.
- AMVF weights as reference favour AMVF by construction.

Result.

- Purity: GNN 91.3%, AMVF 88.4%.

12. Paper figures and Table 1

Comment. Figures 1–7 and Table 1 for the short paper.

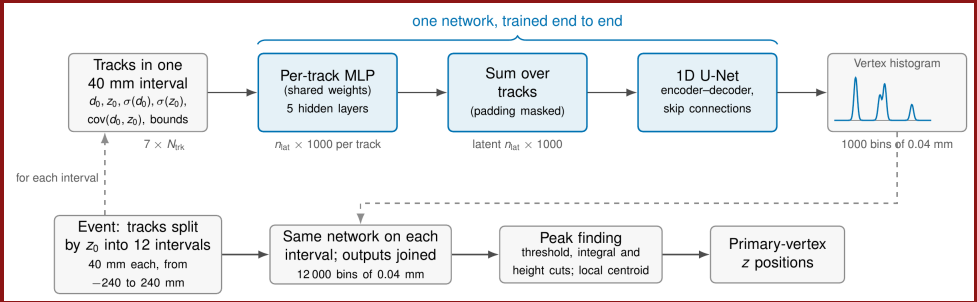
What we did.

- All figures regenerated in ATLAS style from the evaluated samples.
- Captions corrected to the actual samples and μ ranges.

Result.

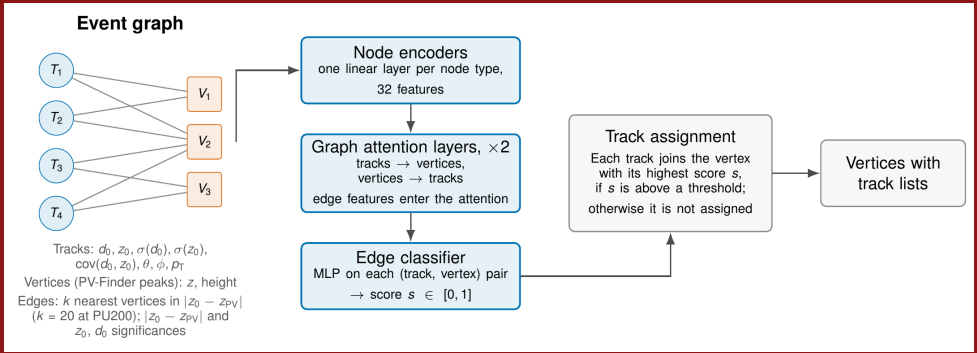
- Fig. 1, 2: PV-Finder and GNN diagrams.
- Fig. 3, 4: HL-LHC vertex categories and Δz at PU200.
- Fig. 5, 6: low pile-up simulation and Run 3 data.
- Fig. 7: timing.
- Table 1: AMVF track requirements, ID Run 3 and ITk Run 4.

Paper Fig. 1: PV-Finder



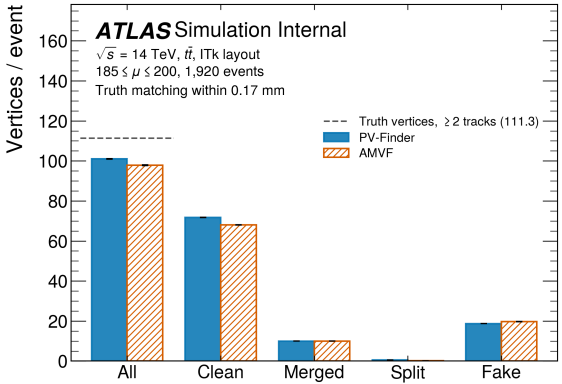
PV-Finder: per-track MLP, latent map, 1-D U-Net, vertex histogram and peak finding.

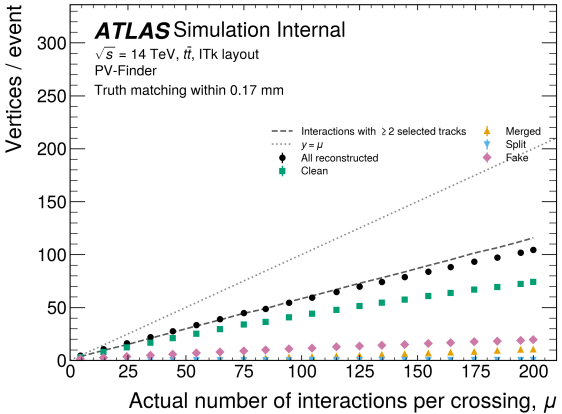
Paper Fig. 2: GNN track-to-vertex association



Graph of tracks and PV-Finder vertices; the GNN scores each track-vertex edge.

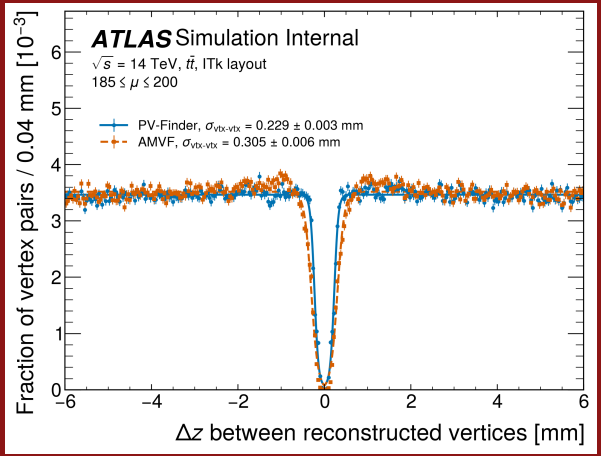
Paper Fig. 3: vertex categories, HL-LHC





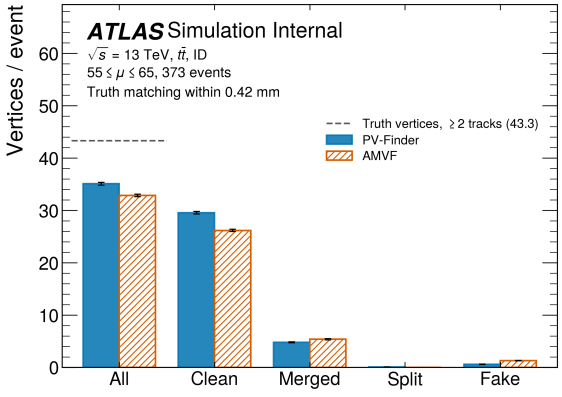
ITk $t\bar{t}$, 14 TeV. (a) Mean vertices per event by category, $185 \leq \mu \leq 200$. (b) Versus μ. Window 0.1694 mm.

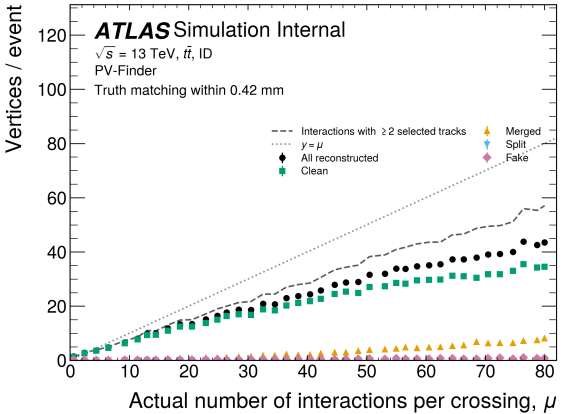
Paper Fig. 4: Δz between vertices, HL-LHC



Distance between reconstructed vertex pairs, PV-Finder and AMVF, $185 \leq \mu \leq 200$. $\sigma_{\text{vtx}-\text{vtx}}$: PV-Finder 0.229 mm, AMVF 0.305 mm.

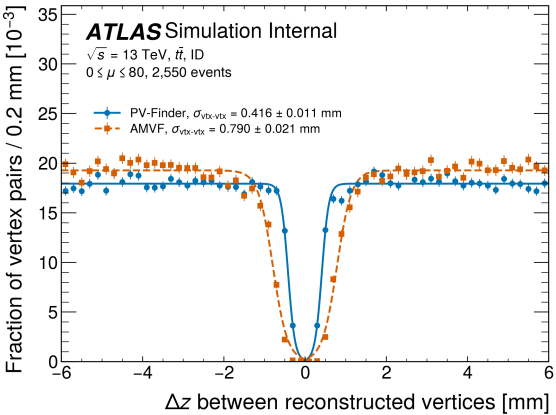
Paper Fig. 5: vertex categories, low pile-up

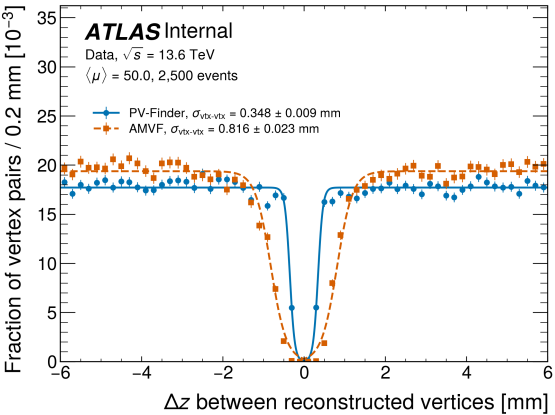




13 TeV $t\bar{t}$ simulation, $\mu = 0$–80, 2 550 events. (a) Categories per event. (b) Versus μ.

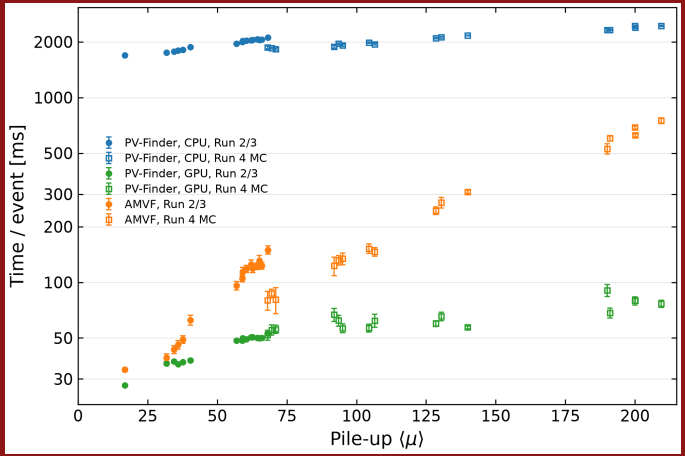
Paper Fig. 6: Δz between vertices, low pile-up





Left: 13 TeV simulation ($\sigma_{\text{vtx}-\text{vtx}}$ 0.416 / 0.790 mm). Right: Run 3 data, $\langle\mu\rangle = 50$ (0.348 / 0.816 mm). PV-Finder / AMVF.

Paper Fig. 7: timing



Per-event wall time, PV-Finder (CPU, GPU) and AMVF. From the note; to be remade with per-event timing.

Paper Table 1: AMVF track requirements

Requirement	ID Run 3	ITk Run 4
Pixel hits	≥ 1	≥ 3
Pixel holes	≤ 1	≤ 1
SCT / strip hits	≥ 4	≥ 0
Silicon hits	≥ 6	≥ 7
p_{T} [MeV]	> 500	> 900
$\lvert d_0 \rvert$ [mm]	< 4.0	< 1.0
$\sigma(d_0)$ [mm]	< 5	< 0.35
$\sigma(z_0 \sin\theta)$ [mm]	< 10	< 2.5

Track requirements applied in AMVF vertex reconstruction, in addition to the track-reconstruction requirements (ITk paper, Table 6).

Next steps

Next

- s8 and s5 (2+2 epochs) held-out results.
- GNN retraining on more data.
- Note edits on a GitLab branch.

Inputs

- Run 3 mc23 $t\bar{t}$ ntuples.
- Per-event Athena timing.
- “Run 2 MC” campaign label.
- Oracle window (0.5 vs 1.0 mm).